Is cosmic fate an inference or an assumption? A locally preregistered fragility audit of the heat-death conclusion with DESI DR2, Pantheon+, and *Planck*

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Abstract

The DESI era has renewed interest in the ultimate fate of the Universe. We ask how much of a fate verdict is constrained by data and how much is induced by assumptions. Using a locally preregistered, version-controlled analysis plan frozen before the first fate calculation, supplemented by explicitly labeled post-hoc audits, we combine DESI DR2 BAO, *Planck* 2018 distance priors, and Pantheon+/SH0ES supernovae, propagate posterior samples through a frozen fate classifier, and audit the verdict along eight declared assumption axes. Baseline CPL gives $P(\textsc{decay}) = 99.82\%$ and $P(\textsc{rip}) = 0.18\%$, with a nested-sampling point estimate of 0.179% (between-seed SD 0.039 percentage points). Within continuous CPL with no point mass at ΛCDM, one hundred fitted-truth ΛCDM mocks yield heat-death-compatible posterior mass consistent with $U(0,1)$; a zero-residual realization gives 52/48. The real-data posterior tail lies below all 100 mocks ($\hat{p} = 0.0099$, 95% Monte Carlo upper bound 0.0295). A two-parameter Wilks test gives $p = 0.024$, but tests a different statistic and is not an independent calibration of that tail. Translation into fate is assumption-dominated: a quintessence prior sets $P(\textsc{rip}) \equiv 0$ at a fit cost of $\Delta\chi^2 = 8.9$, while Bayesian evidence favors ΛCDM ($\ln B = -1.617$, between-seed SD 0.224). Four converged $w(a)$ fits have approximate chain-minimum $\Delta\mathrm{AIC} \in [-0.15, 2.54]$, yet $P(\textsc{rip})$ spans 0.21–49.96% under their native coordinate priors, which induce prior rip mass from 25.8–58.4%. These are sensitivity diagnostics, not a common-prior model comparison: a matched-prior audit returns a structural No-Go because the grammar pushforwards have incompatible singular supports. Finally, the preregistered constraint horizon is *not reached*: BIN4 transports 3.03 nat from its first bin into $a > 1$, while direct observational support ends at $a = 1$. All fits use a *Planck* distance-prior likelihood, so BIN4 remains a geometry-only diagnostic rather than a full-CMB inference. Current fate probabilities therefore combine a pipeline-conditional finite-redshift preference with an unmeasured extrapolation rule; we release the audit framework and pipeline.

I. INTRODUCTION

The DESI era has revived the oldest question in cosmology. Baryon acoustic oscillation measurements from DESI DR2, combined with cosmic microwave background (CMB) and

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Type Ia supernova data, favor solutions in the "thawing" quadrant $w_0 > -1$, $w_a < 0$ [1]. The reported strength is analysis-dependent: official full-likelihood combinations give 2.8–4.2σ, the recalibrated DES-Dovekie sample gives 3.2σ with only weak Bayesian model preference, and a compressed-CMB multi-model analysis reports 1.1–2.3σ [1–6]. Specific parametrizations and physical models have long been translated into future-fate statements, including work designed explicitly to avoid CPL's unbounded future behavior [7–9].

This paper asks a different question. Rather than "what is the fate?", we ask: *of the fate verdict, how much is forced by the data, and how much is donated by the assumptions?* The question has a distinguished qualitative pedigree—Krauss and Turner [10] argued a quarter-century ago that dark energy severs the link between geometry and destiny, and Kallosh and Linde [11] computed observational doomsday bounds for specific potentials. We are unaware of a published study that combines an end-to-end repeated-ΛCDM-mock calibration of a posterior fate-class probability with a finite sensitivity audit over the eight analysis axes declared here. This is the paper's scoped methodological contribution, not a claim that earlier work ignored model dependence.

Our design principle is simple: each main fate conclusion is accompanied by a finite sensitivity audit over declared assumptions. We froze an analysis plan before the first real-data fate computation (Sec. II), validate the pipeline against the official DESI DR2 chains (Sec. III), propagate every posterior sample to $a = 10^4$ through a frozen fate classifier and report the baseline verdict (Secs. IV–V), calibrate the classifier on one hundred mock universes in which the truth is exactly ΛCDM (Sec. VI), and then measure the fate verdict's response along eight axes of assumption variation (Sec. VII): the supernova compilation, the SH0ES anchor, the CMB information content, the dataset combination, the prior on the dark-energy sector, the inference statistic, the $w(a)$ parametrization, and the position of the truth itself.

Three results organize the paper. First, under a regular continuous CPL model and prior, when the truth is exactly ΛCDM the direction-class posterior mass is consistent with $U(0, 1)$ across repeated datasets. This is a conditional boundary-calibration result, not a theorem about model averaging or priors that put discrete mass on ΛCDM. Second, in depth: the real data push the Big Rip probability to 0.18% under CPL, below all one hundred null mocks ($\hat{p} = 0.0099$, 95% upper bound 0.0295). The Wilks likelihood-ratio value [12] $p = 0.024$ points in the same qualitative direction but concerns a different null statistic.

Third, in translation: four converged parametrizations have similar archived chain-minimum AIC values in an exploratory audit, yet the Big Rip probability moves from 0.2% (CPL) to approximately 50% (binned w); under a quintessence prior it is zero before data arrive. A preregistered mean-reverting GP illustrates how a future mean function fixes a fate, but its nonconverged hierarchical fit is excluded from quantitative comparisons. The preregistered KL horizon is not reached; $a = 1$ is instead the boundary between direct observational support and grammar-transported future information (Sec. VII E).

We therefore interpret current fate estimates as conditional predictions whose structural uncertainty is not captured by within-model posterior intervals. The constructive output is a locally preregistered, null-calibrated sensitivity protocol. Within CPL+P1, D0 shows a fitted-truth-mock-calibrated finite-redshift preference; across D1–D4 the posterior anti-rip direction remains sub-percent, although those ablations were not separately null-calibrated. Translation to $a > 1$ additionally depends on the chosen grammar and its induced prior measure.

II. PREREGISTERED DESIGN AND PROVENANCE

The analysis plan specified the model space, priors, dataset combinations, criteria for classifying fates, statistical rules, and pass/fail gates. The protocol was written and frozen in local version control before the first target fate calculation, according to the preserved Git history. Because the repository was made public only afterwards, the original freeze lacks an independently verifiable timestamp. We therefore describe the study as locally preregistered, not as a registry-verified preregistration. Specifically, commit `98abc20` under tag `prereg-v1`, recorded at 10:26 local time on 2026-07-03, is an ancestor of the first real-data fate-result commit recorded at 11:29. The frozen plan is reproduced in English translation in Appendix A; commit ancestry, timestamps, tree identifiers, and the plan checksum are listed in the archived provenance audit, `plan/PREREGISTRATION_PROVENANCE.md`.

More precisely, this is a secondary-data preregistration of the fate-translation audit. The datasets and validation strategy were already known, but the core classifier, audit axes, gates, and two reportable outcome forms were frozen before the first target fate calculation. It is not a claim of preregistration before data access or of an independently timestamped public registration.

Departures from the frozen plan are recorded in an append-only log (Appendix B) stating what changed, why, and when relative to data contact. Three amendments to the frozen core occurred: A-001 recalibrated the pipeline-validation benchmark after we established that a compressed-CMB likelihood cannot reproduce the full-likelihood exclusion significance (Sec. III); A-002 replaced a mock-recovery criterion that is mathematically unsatisfiable for a truth on a fate boundary (Sec. VI); and A-003 added an analytic fate branch after the frozen OTHER-budget audit was triggered by bounded-future parametrizations (Sec. VII D). In each case the trigger, argument, and sign-off are logged. Four later records are explicitly post-hoc: A-004 tests whether the grammar-specific priors can be matched by importance reweighting and returns a global No-Go; A-005 corrects the finite-limit fate semantics without altering CPL-based results; A-006 adds an early-dark-energy hard gate and records the full-Planck No-Go for the current background-only BIN4 implementation; and A-007 documents the model-specific translation of P1's early-matter-domination intent without retroactively treating that implementation detail as explicitly preregistered.

The data also predate the plan: DESI DR2 constraints were public before the freeze. The version-controlled record documents analysis-choice discipline—a declared classifier, gates, fragility metrics, and two equally reportable outcome forms—rather than prediction novelty. The published w_0–w_a constraint is used only as a Gate-1 validation target. Six analytical checks for the null campaign were recorded after approximately 17 of 100 mocks and before the remaining campaign completed; they are prospective checks for the remaining runs, not pre-analysis registrations. The zero-residual prediction was recorded before that run (Appendix D). This discipline is weaker than survey blinding, which can shield catalogs or data vectors from the analyst [13, 14]; it makes the workflow auditable but does not eliminate all analyst degrees of freedom. The methodological contribution is the resulting transparent sensitivity audit, not an external-registry claim.

III. DATA, LIKELIHOODS, AND VALIDATION

We use three publicly released datasets: DESI DR2 BAO compressed measurements (13 points over seven redshift bins, $0.295 \leq z \leq 2.33$) [1]; the Pantheon+ Type Ia supernova compilation with full stat+sys covariance, with and without the SH0ES Cepheid anchor [15–17]; and *Planck* 2018 information [18] compressed into the distance priors $(R, \ell_A, \Omega_b h^2)$ of

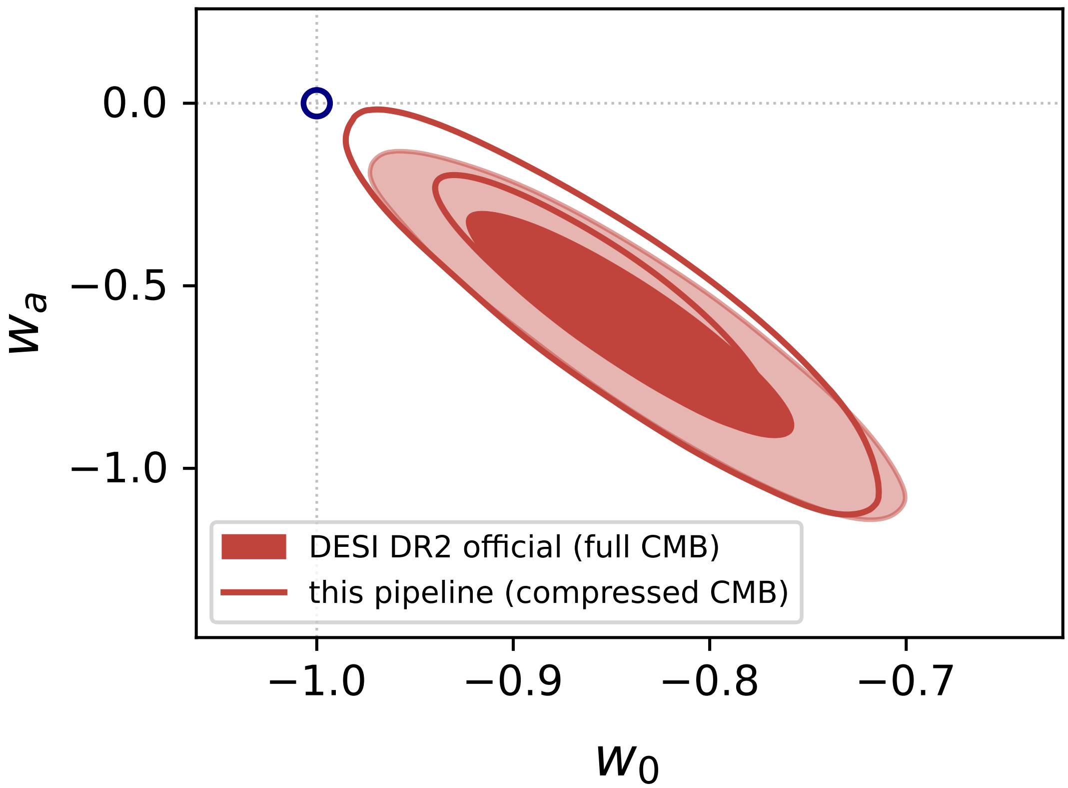

FIG. 1. Gate-1 chain-level validation: this pipeline (compressed CMB, open contours) versus the official DESI DR2 chains (full CMB + lensing, filled). All marginal shifts $\leq 0.33\sigma$. Here and throughout, contour pairs show the 68% and 95% credible regions.

Ref. [19], validated there for CPL-type models. Files, versions, and checksums are registered in the repository manifest (Appendix F).

Two validation results define what the pipeline can claim. First, against the official DESI DR2 w_0w_aCDM chains, our marginals agree to $\leq 0.33\sigma$ in every parameter, with Ω_m and H_0 agreeing to $< 0.1\sigma$ (Fig. 1). Second, the compressed CMB prior carries a known, stable information deficit: our ΛCDM exclusion is 2.25σ ($\Delta\chi^2 = 7.44$) where the full-likelihood value is 2.8σ, and the same $\approx 0.5\sigma$ discount recurs across all three supernova compilations (Sec. VII A), consistent with independent compressed-likelihood analyses [3]. Every significance below inherits the compressed-CMB condition; we price it rather than hide it. One procedural lesson is worth recording: compressed priors must be predicted with the same z_*/r_s conventions used at extraction— mixing conventions shifted ℓ_A by 3σ of pure artifact in an early pipeline version, caught by a known-answer test before any science ran.

IV. FATE TAXONOMY AND CLASSIFIER

Each posterior sample defines a background $H^2(a)/H_0^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE} f_{\rm DE}(a)$, integrated from $a = 1$ to $a_{\max} = 10^4$ and classified with frozen criteria, in priority order: CRUNCH ($H^2 \leq 0$ reached); RIP (finite-time singularity [20]: convergent $\int da/aH$; for parametrizations whose $w(a)$ has a finite limit w_∞, A-003 introduced an analytic branch and A-005 corrects its physical boundary to $w_\infty < -1$); DS (exact de Sitter asymptotics, $w_\infty = -1$ with persistent dark energy, in the finite-limit branch); DECAY (dark-energy dilution); OTHER (unclassifiable—audited under a frozen 1% budget whose breach triggers the amendment process, as occurred for A-003). Every class carries a boundary flag: numerical-branch samples that flip when thresholds are doubled or halved, and finite-limit samples satisfying $|w_\infty + 1| \leq 0.01$, are reported as boundary-adjacent without changing their physical label.

For this audit, DS and DECAY are grouped into a coarse heat-death-compatible macroclass because both avoid a finite-time singularity and expand forever in the flat backgrounds considered here. This operational grouping does not assert thermodynamic equivalence: horizon structure, temperature floors, and accessible free energy can differ between the two. Within flat CPL backgrounds the only non-heat-death branch is the Big Rip. The classifier passes a ten-case unit suite spanning all classes, boundary behavior, and two documented numerical edge cases (Appendix C).

V. BASELINE FATE PROBABILITIES

Figure 2 shows the D0 (DESI DR2 + CMB prior + Pantheon+SH0ES) CPL [21, 22] posterior—$w_0 = -0.884 \pm 0.057$, $w_a = -0.62^{+0.25}_{-0.22}$, $\Omega_m = 0.2997 \pm 0.0050$, $H_0 = 69.00 \pm 0.54$ km s^{-1} Mpc^{-1} (four chains, $R - 1 = 0.0033$)— drawn over the fate partition of the (w_0, w_a) plane. The posterior presses against the zero-width RIP/DECAY boundary at $w_a = 0$; the consequences of that geometry occupy the next section.

Propagating every sample through the classifier yields Table I: P(DECAY) $= 99.82\% \pm 0.06\%$, P(RIP) $= 0.18\% \pm 0.06\%$ (batch-means errors), zero OTHER, zero boundary flags. Under the frozen rule that sub-percent tails require a second sampling calculation, nested sampling gives a consistent weighted point estimate, P(RIP) $= 0.179\%$. Across three inde-

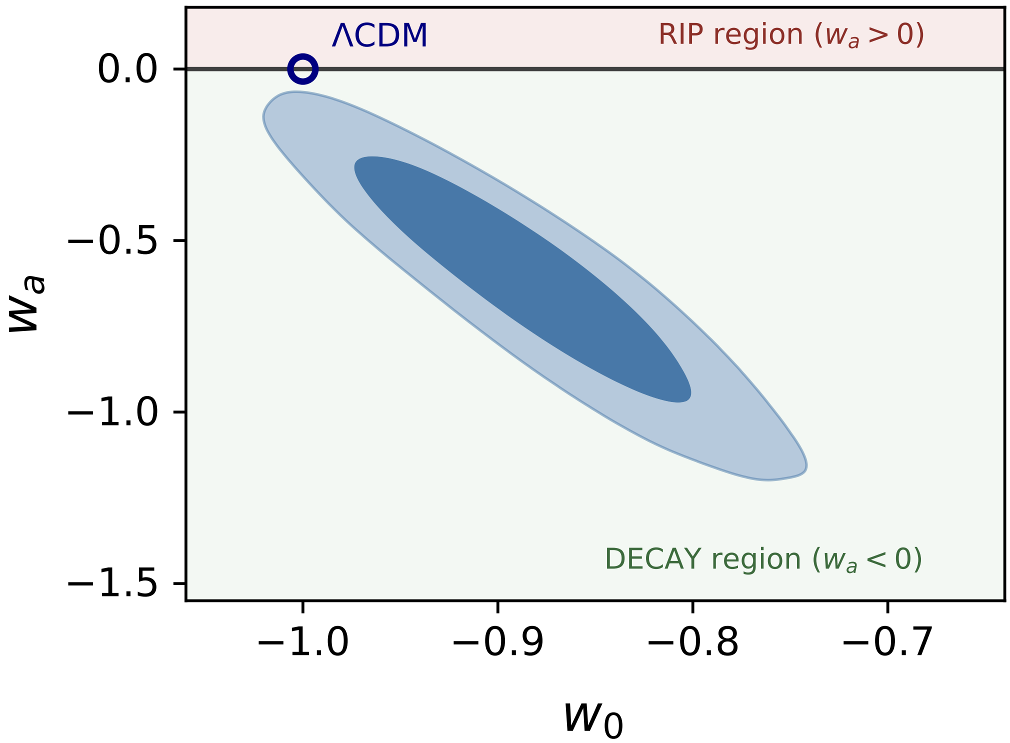

FIG. 2. D0+CPL+P1 posterior over the fate partition. The fate boundary $w_a = 0$ has zero width; ΛCDM sits exactly on it.

TABLE I. Fate probabilities, D0+CPL+P1. *Caution: the direction component of this verdict is uninformative under the null (Sec. VI).*

Class	P	thermodynamic class
DECAY	$99.82\% \pm 0.06\%$	heat-death-compatible
RIP	$0.18\% \pm 0.06\%$ (nested 0.179%)	non-heat-death
DS	0 (grammar artifact; Sec. VII D)	heat-death-compatible
OTHER/boundary	0 / 0	—

pendent seeds, the between-seed standard deviation is 0.039 percentage points and the range is 0.135–0.209%. The previously quoted 0.038% binomial standard error describes only a fixed-seed equal-weight resample; it is not run-to-run nested-sampling uncertainty and is therefore not used as an error bar. The weighted nested OTHER mass is 0.0093%, below the registered budget.

Two seeds of unease belong here. First, for CPL against ΛCDM, the three-seed mean is $\ln B = -1.617$. The between-seed SD is 0.224, with range $[-1.865, -1.431]$. The three per-run internal dynesty errors span 0.313– 0.314. Bayesian evidence *favors* ΛCDM while $\Delta\chi^2$ disfavors it at 2.25σ—a live Lindley phenomenon [23], developed as a fragility axis in Sec. VII C. Second, $P(\text{DS}) = 0$ exactly: strict de Sitter asymptotics is a measure-zero locus of CPL parameter space, so this zero was decided by the parametrization before any photon arrived. Before interpreting Table I at all, however, one must ask what such a table looks like when the truth is exactly ΛCDM—the subject of the next section.

VI. NULL CALIBRATION

We are unaware of a published repeated-ΛCDM-mock calibration of a posterior cosmic-fate class probability using the same end-to-end inference pipeline. We generated one hundred mock datasets whose truth is the ΛCDM best fit to D0, drawing noise from the exact likelihood covariances (SN 1657×1657; BAO 13×13; CMB prior 3×3), plus one zero-residual realization for which all three χ^2 vanish at the truth—a closed-loop validation of the mock machinery. Each mock was processed by the full CPL+P1 pipeline.

Because the ΛCDM truth sits on the zero-width CPL fate boundary, local regular asymptotics predict the heat-death-compatible posterior mass to approach $U(0, 1)$ across repeated realizations: the local likelihood must be calibrated, the boundary locally linear, and the prior absolutely continuous with no point mass on ΛCDM. Under those conditions a direction verdict is correct in $\sim 50\%$ of mocks. This is a property of the specified decision problem, not a statement that pipeline quality is irrelevant and not a universal limit on Bayesian procedures. Spike-and-slab priors, explicit ΛCDM model probability, curved boundaries, or misspecified posterior widths change the result (Appendix E). These predictions were logged mid-campaign (at ∼17 of 100 mocks), before the campaign's completion and before the zero-residual run. Results (Fig. 3): accuracy 50/100 (68% CI $[45, 55]$); KS uniformity $p = 0.25$; mean 0.525 ± 0.030; OTHER and boundary budgets at the 10^{-3} level. (A hundred-realization KS test has limited power against mild distortions, which is why the registered predictions also pinned the mean, the accuracy count, and the budgets—all separately satisfied, while retaining limited power against mild departures.) The zero-residual mock returns 52/48. Its data vector equals the truth, but its likelihood covariance remains finite; it is a closed-

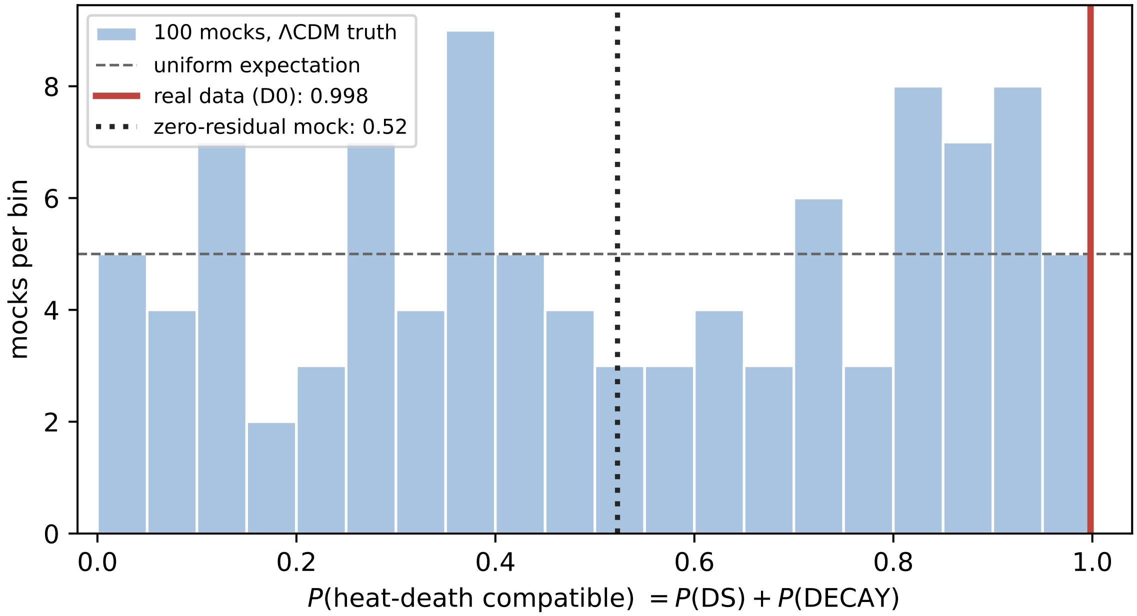

FIG. 3. Null calibration of the fate verdict. Under ΛCDM truth the heat-death-compatible posterior mass is consistent with $U(0,1)$ (KS $p = 0.25$) for continuous CPL+P1; the zero-residual mock returns 52/48. The real-data value (red) lies beyond every mock realization.

loop diagnostic, not a zero-uncertainty experiment. The result isolates the role of boundary geometry within this continuous CPL analysis.

The null distribution also supplies the calibrated reading of the real data. Across one hundred ΛCDM universes the lowest Big Rip probability produced by noise is 1.52%; the real Universe sits at 0.18%—eight times deeper, below every realization. The conditional parametric-bootstrap rank at this fitted truth is $\hat{p} = (0+1)/(100+1) = 0.0099$, with a 95% upper bound of 0.0295. The asymptotic Wilks route gives $p = 0.024$, but the statistics are not equivalent: the mock campaign ranks the one-sided posterior mass $P(w_a > 0 \mid D)$ within a CPL pipeline, whereas Wilks tests the point $(-1, 0)$ against a two-parameter CPL alternative. Their numerical agreement is a useful qualitative cross-check, not two independent measurements of the same tail. The correct summary of Table I is therefore two-layered: *the CPL direction probability has the registered uniform null distribution, while the real data lie unusually deep in its anti-rip tail and also disfavor point-*ΛCDM *by a distinct likelihood-ratio test.* What that signal does and does not imply about the actual fate is the subject of the fragility matrix.

TABLE II. Leave-one-out (CPL+P1 fixed). Per the frozen sub-percent rule, the sub-percent entries were recomputed with nested sampling. The weighted point estimates are consistent in direction with the MCMC values; for D3 the original weights resolve a small positive tail that the equal-weight diagnostic rounds to zero. Binomial errors stored with the equal-weight resamples are numerical resampling diagnostics, not repeated nested-run uncertainties (per-combination records in the repository).

Combo	removed	w_0	w_a	$P(\text{RIP})$
D0	—	-0.884 ± 0.057	-0.62	0.18%
D1	SH0ES	-0.848 ± 0.055	-0.55	0.376%
D2	CMB	-0.853 ± 0.067	-1.06	0.067%
D3	BAO	-0.711 ± 0.083	-1.64	0.0008%
D4	SN	-0.43 ± 0.22	-1.72	0.189%

VII. THE FRAGILITY MATRIX

A. Data axis

Table II and Fig. 4 show the leave-one-out response. The thawing direction ($w_0 > -1$, $w_a < 0$) survives every ablation—including removal of the CMB (D2; the CMB prior is replaced by a BBN baryon-density prior [24]), which *deepens* rather than erases the preference, a point against the hypothesis that the evidence is purely CMB–low-z tension. The magnitude, by contrast, is fragile: w_a swings threefold (-0.55 to -1.72), and the full combination is more moderate than any probe-removal subset (D2–D4)—the widely quoted $w_a \approx -0.6$ reflects partially offsetting dataset pulls. Swapping the supernova compilation (same model, same priors, same BAO and CMB) moves the ΛCDM exclusion from 2.25σ (Pantheon+) through 2.81σ (DES-Dovekie) to 3.27σ (Union3 [25]), reproducing the literature's significance ladder [4, 5] inside a single pipeline; the SH0ES anchor alone moves H_0 by $+1.35\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and w_0 by -0.034 —Hubble-tension leakage into the dark-energy sector. Throughout, $P(\text{RIP})$ stays within 0.0008–0.376%: the direction-level conclusion remains sub-percent for every combination, but the numerical tail spans 2.665 dex and is therefore not robust in a multiplicative sense. Grammar dependence is larger still (Sec. VII D).

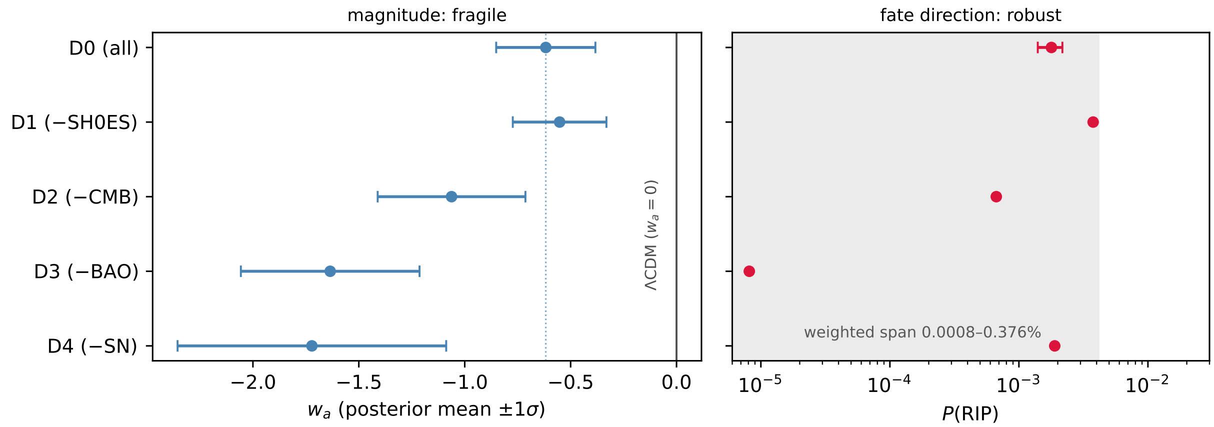

FIG. 4. Leave-one-out forest (CPL+P1 fixed). Left: the w_a magnitude swings threefold across dataset ablations, and the full combination (dotted line) is more moderate than any probe-removal subset (D2–D4). Right: $P(\textsc{rip})$ stays within 0.0008–0.376% throughout (shaded); the D3 entry is the small weighted tail retained by the original nested weights even though its fixed-seed equal-weight diagnostic contains zero RIP draws.

B. Prior axis

Under the physical priors the question dissolves before data arrive: P2 (pointwise $w(a) \geq -1$, quintessence) and P3 (P2 + thawing) force $P(\textsc{rip}) = 0$ *identically*—reported, per the frozen plan, as "excluded by prior." The posterior is pinned into a sliver against the $w = -1$ wall ($w_0 = -0.975$, $w_a = -0.015$), and the wall costs the fit $\Delta\chi^2 = +8.9$ relative to P1: a clean measure of the current tension between the data and the quintessence hypothesis. P3 adds nothing to P2 on these data—its extra thawing assumptions are redundant once the pointwise bound binds. The prior axis reading is therefore maximal: $\log_{10} P(\textsc{rip})$ spans from -2.7 to $-\infty$.

C. Inference-statistic axis

The same data, model, and priors yield the fit improvement $\Delta\chi^2 \equiv \chi^2_{\Lambda\mathrm{CDM}} - \chi^2_{\mathrm{CPL}} = 7.44$ (Wilks $p = 0.024$, disfavoring ΛCDM) and a three-seed $\ln B = -1.617$ (between-seed SD 0.224; Jeffreys-scale support *for* ΛCDM). Nothing is inconsistent: the two statistics ask different questions, and the Occam penalty of the 15.5-unit (w_0, w_a) prior area dominates the fit improvement. But the practical consequence is stark—*whether "the evidence favors dynamical dark energy" is true depends on which inference statistic one grants authority*, and the Bayes factor's first-order dependence on prior volume makes it a fragility axis of its own, rarely reported as such. The registered information criteria split along the same fault line: $\Delta\text{AIC} = -\Delta\chi^2 + 2\Delta k = -3.4$ sides with the fit, while $\Delta\text{BIC} = -\Delta\chi^2 + \Delta k \ln N = +7.4$ sides with the evidence ($\Delta k = 2$, $N = 1673$ data points; negative values favor CPL)—the direction flip is reproduced within the information-criterion family itself.

An explicitly model-averaged reading makes the conditioning visible. As an *exploratory, post-hoc* diagnostic, assign equal prior odds to the two-model set $\{\Lambda\text{CDM}, \text{CPL}\}$. The measured Bayes factor then gives $P(\Lambda\text{CDM} \mid D) = 83.4\%$ and $P(\text{CPL} \mid D) = 16.6\%$; because exact ΛCDM is heat-death-compatible, the corresponding model-averaged compatibility is 99.97%. These values are not privileged—they change with the model set and model prior odds—but they demonstrate why the continuous-CPL 50/50 boundary result cannot be promoted to a universal Bayesian claim.

D. Parametrization axis

Table IV and Fig. 5 are the paper's central result, but two comparisons must be separated. First, what do the archived chains say about in-window fit? An exploratory post-hoc audit (Table III) finds approximate minimum likelihood ΔAIC between −0.15 and +2.54 relative to CPL. These are chain minima, not dedicated optimizations or evidences; they support only the descriptive statement that the archived chain-minimum AIC values are similar. The literature establishes related in-window degeneracies for overlapping two-parameter families [26], not for this entire set and not for our BIN4 implementation.

Second, the four fits do *not* have identical priors. Their coordinate dimensions, early-matter-domination constraints, and induced measures on functions and fates differ. A deterministic P1 prior audit makes the baseline explicit: prior $P(\text{RIP})$ is 25.8% (CPL), 40.0% (JBP), 58.4% (BA), and 50.6% (BIN4). The likelihood strongly suppresses the rip within every fitted grammar, but the residual posterior remains grammar-dependent. The divergent-future CPL and JBP families deliver near-certain DECAY; bounded-future BA retains a small RIP component; and frozen-future BIN4 returns 49.96% RIP and 50.04% DECAY, with strict DS mass zero. Its fate-controlling bin is measured as $w_1 = -1.000 \pm 0.044$. Under A-005,

TABLE III. Exploratory in-window fit audit. Minima are the lowest likelihood χ^2 values in the archived post-burn-in chains; common parameters cancel in ΔAIC. This is not a model-evidence calculation.

Grammar	$\chi^2_{\text{chain,min}}$	$\Delta\chi^2$	ΔAIC	$R-1$
CPL	1489.534	0.000	0.000	0.0063
JBP	1492.075	2.541	2.541	0.0093
BA	1491.523	1.989	1.989	0.0026
BIN4	1485.386	-4.149	-0.149	0.0046

17.6% of BIN4 posterior mass is reported as boundary-adjacent without changing its exact fate label. The binned reconstruction also relocates the finite-redshift signal: the low-z bin is ΛCDM-consistent and the deviation lies at $z \in [0.7, 1.5]$ ($w_3 = -1.53 \pm 0.20$). Thus CPL's $w_0 \approx -0.88$ is partly the transport of a mid-z signal to $z = 0$ by a linear grammar.

A post-hoc matched-prior audit stops before importance weighting. On the declared seven-point $w(a)$ grid, the CPL, JBP, and BA pushforwards occupy different two-dimensional supports and BIN4 occupies a three-dimensional support: the grid reaches only $z = 1$, so its summary is $(w_3, w_2, w_1, w_1, w_1, w_1, w_1)$ and contains no w_4 coordinate. Their common intersection contains only constant histories and has probability zero under every continuous native prior. A nondegenerate common target is therefore not dominated by all four pushforwards; diagonal covariance regularization would manufacture overlap rather than identify a matched measure. A-004 accordingly reports a global No-Go and withholds all matched fate probabilities.

The registered Gaussian-process branch is reported differently. Its hierarchical chain stopped with $R-1 = 0.232$, failing Gate 3; moreover the box-truncated hierarchical density was not normalized as a function of its hyperparameters. We therefore remove all GP node, kernel-sensitivity, and fit-comparison claims. One exact construction remains valid without those claims: assigning the conditional-mean future of a stationary GP with mean $w = -1$ makes $w(a \to \infty) = -1$ for every draw, so the asymptotic classifier returns DS by construction. This is a transparent counterexample showing that a future mean function can fix the fate before the likelihood is evaluated; it is not a fifth converged fit. Among the four fitted grammars, $P(\text{RIP})$ spans 0.21–49.96% and heat-death compatibility 50.04–99.79%.

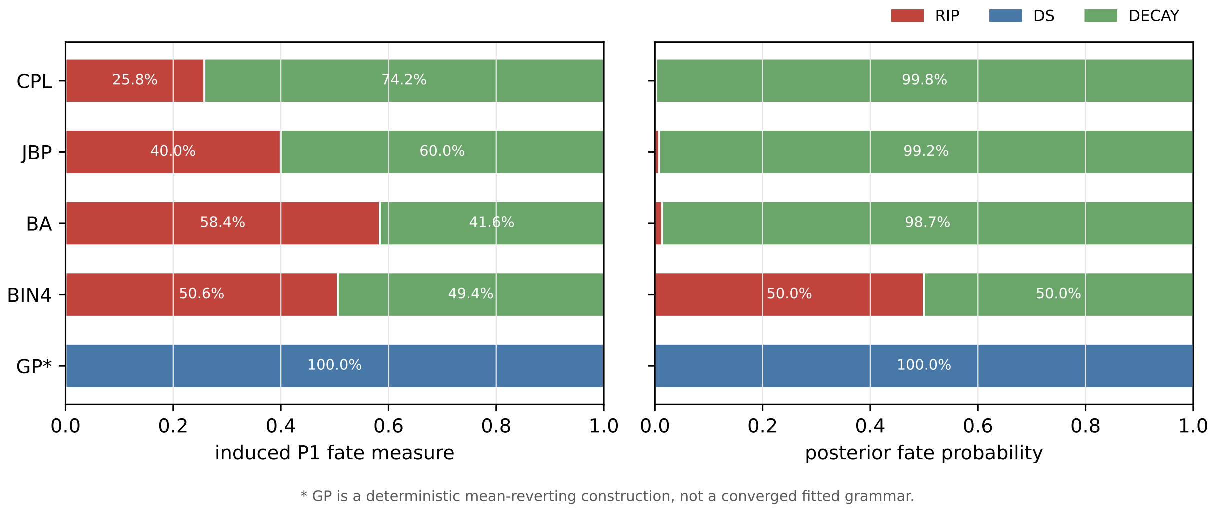

FIG. 5. Prior and posterior fate composition under the grammar-specific P1 measures (A-005 corrected semantics). Flat coordinate priors do not induce a common fate measure. GP* is the deterministic mean-reverting construction, not a converged fitted grammar.

TABLE IV. Grammar-specific induced prior fate mass and posterior fate probabilities (D0 fixed; A-005 corrected semantics). The CPL posterior row uses the generic cross-grammar background machinery and agrees with the baseline within Monte Carlo error. GP* is a construction only.

Grammar	prior $P(\text{RIP})$	posterior $P(\text{RIP})$	posterior $P(\text{DECAY})$	posterior $P(\text{DS})$	posterior $P(\text{heat-death})$
CPL	25.8%	0.21%	99.79%	0	99.8%
JBP	40.0%	0.79%	99.21%	0	99.2%
BA	58.4%	1.34%	98.66%	0	98.66%
BIN4	50.6%	49.96%	50.04%	0	50.04%
GP*	construction: 0	0	0	100%	100%

Adding the construction extends the latter endpoint to 100%.

E. Constraint horizon

Where do the data stop speaking? In the binned grammar the posterior-to-prior information $D_{\text{KL}}[p(w(a))\|\pi]$ is 3.03, 2.31, 1.55, and 0.24 nat in the four bins (Fig. 6)—the frozen 0.1-nat horizon is never crossed inside the observed window: the compressed CMB lever arm constrains the highest-redshift bin weakly but above threshold. The registered metric must then be applied to the registered future rule. BIN4 freezes $w(a > 1) = w_1$, so both its posterior and prior are inherited from w_1 and the future KL remains 3.03 nat. Thus a_h is *not reached*; reporting $a_h = 1$ would be inconsistent with the registered metric.

A separate and scientifically important statement does hold: $a = 1$ is the boundary of *direct observational support*. The nonzero KL plotted for $a > 1$ was learned about w_1 at $a \leq 1$ and transported into the future by the BIN4 freezing rule. CPL transports information differently. Thus future fate is not supported by direct likelihood evaluations at $a > 1$, even though the posterior-to-prior KL of the extrapolated function need not vanish.

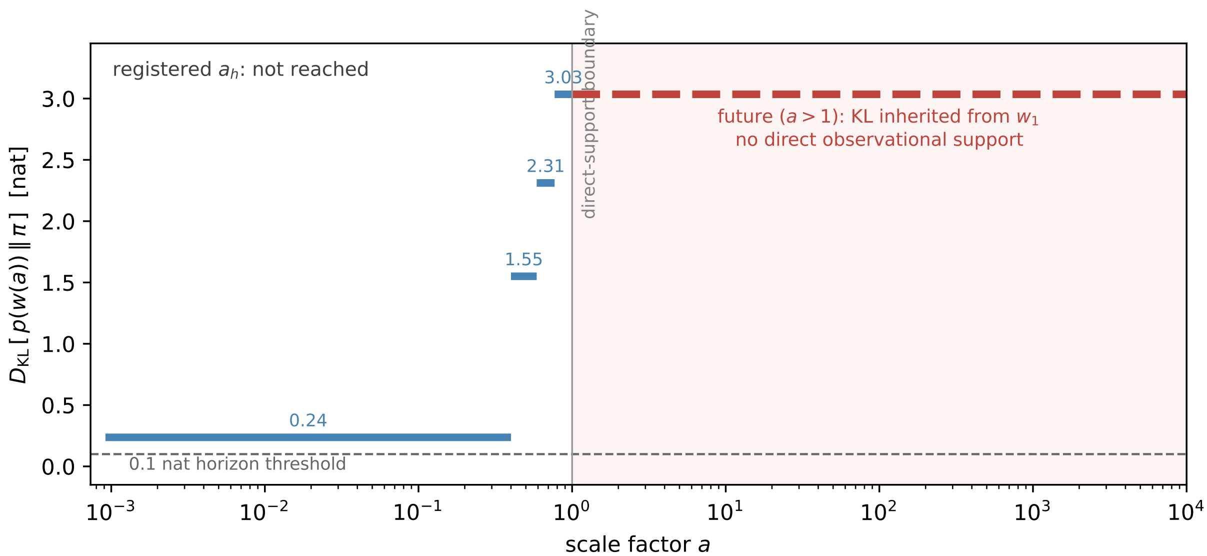

FIG. 6. Constraint horizon (BIN4): posterior-to-prior information in $w(a)$ per bin. The dashed future segment is the w_1 KL inherited by the BIN4 freezing rule, not direct future information. The frozen 0.1-nat threshold is never crossed, so the registered result is a_h: not reached. The vertical line at $a = 1$ marks the distinct boundary of direct observational support.

F. Synthesis

The three registered log-span metrics are fulfilled explicitly in Table VI; they are not replaced by the broader eight-axis narrative matrix. Exact zeros imposed by P2/P3 and by the GP construction make the corresponding RIP spans infinite. No original-weight $F_{\rm data}$ point is zero in v1.2: D3 retains a small positive tail even though its equal-weight diagnostic contains zero RIP draws. The registered rule remains that a finite-run zero would not be

TABLE V. The eight-axis fragility matrix. Each row varies one audit package or axis and reports the response of the fate verdict. The grammar row necessarily couples functional form, native dimension, and its registered coordinate prior.

Axis	knob	response	verdict
SN compilation	Pantheon+/Dovekie/Union3	ΛCDM exclusion $2.25-3.27\sigma$	magnitude fragile
SH0ES anchor	in/out	H_0 $+1.35$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; w_0 -0.034	leakage; direction stable within CPL
CMB information	CPL compressed/full benchmark	stable -0.5σ discount	BIN4 remains compressed-only
Dataset ablation	D1–D4	$w_a \in [-1.72, -0.55]$; $P(\textsc{rip})$ $0.0008-0.376\%$	direction stable; tail 2.665 dex
Prior	P1/P2/P3	$P(\textsc{rip})$: $0.18\% \to 0$ (excluded)	prior decides
Statistic	$\Delta\chi^2$ vs $\ln B$	2.25σ against vs. $\ln B = -1.617$ (CPL/ΛCDM)	conditioning flips
Grammar	four fits + GP construction	$P(\textsc{rip})$ $0.21 \to 49.96\%$ in fits	dominant axis
Truth position	ΛCDM in continuous CPL	verdict $\sim U(0,1)$; zero-residual 52/48	conditional null law

TABLE VI. Registered fragility measures, in dex. The fitted-four row is a diagnostic subset of the registered F_{param} set, which also includes the deterministic GP construction.

Axis	$P(\textsc{rip})$ span	$P(\text{heat})$ span
F_{prior}	∞	0.002
F_{param}, registered	∞	0.301
F_{param}, fitted four	2.380	0.300
F_{data}	2.665	0.002

promoted to a mathematical zero.

Table V assembles the audit. For D0 within CPL+P1, the posterior-tail mock rank and the non-equivalent Wilks statistic both lie near the few-percent level. Across D1–D4, the anti-rip posterior direction remains sub-percent within the same CPL pipeline, but those variants were not separately null-calibrated and the tail magnitude spans 2.665 dex. Quantities specifically about fate—which endpoint, with what probability—also respond strongly to theory-side choices that the current likelihood does not directly constrain.

VIII. DISCUSSION

Three layers. The audit resolves the fate question into three layers with different epistemic status. (i) *Direction*: within the continuous CPL decision problem, the fate-side posterior mass approaches a uniform null distribution when the truth lies on the local ΛCDM boundary. This statement does not transfer automatically to BIN4 or to explicit model averaging. (ii) *Depth*: within CPL+P1 and the adopted compressed-CMB likelihood, the D0 posterior lies in the anti-rip tail of the fitted-truth mock distribution ($\hat{p} = 0.0099$, 95% Monte Carlo upper bound 0.0295); the non-equivalent Wilks test gives $p = 0.024$. This is a pipeline-conditional finite-redshift preference, not a calibration of data systematics, likelihood compression, or cross-model uncertainty. Across D1–D4 the anti-rip posterior direction remains sub-percent within CPL+P1, but these variants were not separately null-calibrated. Separately, BIN4 places its largest in-window departure at $z \in [0.7, 1.5]$. (iii) *Translation*: converting finite-redshift information into a fate probability additionally requires an extrapolation rule and induced prior measure that observations do not directly constrain.

Why the fragility is structural, not accidental. The fate map is a *discontinuous* functional of parameters that observation constrains only *continuously*: RIP and DECAY are separated by a zero-width boundary, and ΛCDM sits on it. Under regular continuous-model asymptotics, repeated-data class probabilities remain uniform when the truth is exactly on a locally linear boundary, even as the likelihood narrows. This statement is conditional: a discrete ΛCDM component, model averaging, nonregular priors, or a physically derived theory changes the inferential object. In that qualified sense the result is a quantitative instance of the underdetermination of dark energy articulated in Refs. [27, 28], not a no-go theorem for every future Bayesian procedure.

What would change the situation. More precise distance data can narrow the finite-redshift function and may reveal that the truth is off the boundary, but observations made today cannot directly sample $a > 1$. The registered KL horizon is not reached; the relevant limitation is instead the direct-support boundary at $a = 1$. Extrapolation is licensed by trusted theory, not by fit quality alone. CPL and its cousins are more than raw curve fitting—they are compressions calibrated against quintessence-class dynamics inside the window [29–31]—but a calibrated compression is still not a theory: the calibration certifies fidelity to a model class over the fitted range, and extending that fidelity to $a > 1$ is precisely the adoption of the class as a prior—an axis this audit prices (Sec. VII B). A physically derived dark-energy model (a specific potential, a symmetry argument) would convert fate back into a derivation—then data would test the theory inside the window and the theory would carry the license beyond it. Until then, fate probabilities remain conditional predictions of the adopted model class.

Relation to the recent fate literature. Papers reporting P(fate) have always been conditional on a parametrization or physical model. Earlier work explicitly replaced CPL with a bounded future parametrization, and recent studies examine multiple parametrizations or specific scalar-field scenarios [7–9, 26]. These are scientifically meaningful conditional calculations. Our narrower contribution is to report the induced prior fate mass, cross-grammar sensitivity, and fitted-truth null calibration in one audit. Table V is therefore a finite, non-exhaustive sensitivity budget, not a universal model-uncertainty interval. The framework—local preregistration, null calibration, and declared assumption axes—can be adapted to other inferences whose conclusions lie beyond their data window.

Limitations. All significances inherit the compressed-CMB condition (priced at $\approx 0.5\sigma$; A-001). That condition is mildest for CPL-like histories, for which the distance priors were validated [19]; for the binned grammar, whose high-z bin can depart further from Λ, extraction-side validity is an additional inherited assumption. A-006 replaces the insufficient $w_4 < 0$ claim with the hard gate $\rho_{\mathrm{DE}}/\rho_m(z = 1059) < 0.01$; the archived posterior has zero weighted violating mass and a maximum sampled ratio 8.62×10^{-5}. A full-Planck BIN4 run is nevertheless a No-Go in the present geometry-only code because it supplies no CMB spectra or perturbation evolution. BIN4 results are thus compressed-CMB-conditioned. Growth data could shift the window-interior signal. The four fitted grammars are not a model average and their induced priors differ; the prior audit exposes but does not remove that dependence. The A-004 post-hoc audit further shows that this dependence cannot be removed by importance reweighting on the declared seven-point summary, because the grammar pushforwards have incompatible singular supports. The GP is a construction rather than a fit, because the hierarchical chain failed the convergence gate and its truncated hyperprior requires proper normalization before quantitative reuse. The grammar family is not exhaustive—further parametrizations and reconstructions are in active use [26]—so the measured span is not a universal bound. The work was performed by a single author with AI assistance under the verification protocol described; residual unknown-unknowns are bounded only by the audit trail, which is public.

IX. CONCLUSIONS

Within CPL+P1 and the adopted compressed-CMB likelihood, the data exhibit a pipeline-conditional finite-redshift preference: the real CPL tail lies below all one hundred fitted-truth ΛCDM mocks, and a distinct Wilks test gives a similar few-percent scale. Translating that preference into fate remains conditional. A quintessence prior excludes the rip before data; four converged native-prior grammars with similar archived chain-minimum AIC values move posterior $P(\textsc{rip})$ from 0.21% to 49.96% against induced prior values of 25.8–58.4%. These values diagnose sensitivity and are not a common-prior model comparison. A mean-reverting GP construction fixes DS without qualifying as a converged fit. The preregistered a_h is not reached, while $a = 1$ marks the boundary of direct observational support. Current data therefore support a finite-redshift dynamical-dark-energy conditional, not a model-independent cosmic fate. The audit framework makes the conditioning and its error budget explicit.

ACKNOWLEDGMENTS

This work used `CAMB` [32], `cobaya` [33], `GetDist` [34], and `dynesty` [35]; we thank their authors, and the DESI collaboration for the public release of the DR2 cosmology chains used in validation. Substantial portions of the pipeline implementation, literature triage, and manuscript drafting were performed with an AI assistant (Claude, Anthropic) under the version-controlled verification protocol described in Sec. II; all analysis choices, amendment sign-offs, and final claims are the author's responsibility.

Appendix A: Frozen analysis plan

The plan below is an English rendering of the locally frozen document under the historical tag `prereg-v1` (drafted 2026-07-02 and tagged 2026-07-03 before the first real-data fate calculation). As explained in Sec. II, this was a local, version-controlled preregistration; it was not deposited in a third-party registry at freeze time, and the later public repository does not independently establish the original timestamp. The frozen original is archived in the repository and takes precedence in any discrepancy. Amendments are in Appendix B.

Research question and pre-declared outcomes. The deliverable is a set of fragility measures, not a fate discovery. Two outcome forms were declared in advance as equally publishable: (A) fate probabilities are data-dominated (variation across priors/parametrizations below one order of magnitude), or (B) fate probabilities are assumption-dominated (variation of one order of magnitude or more, or a fate decision falling outside the constraint horizon). The plan records no preference between them.

Model. Background expansion only (no perturbations; growth data out of scope): $H^2(a)/H_0^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{\rm DE} f_{\rm DE}(a)$, with Ω_r fixed from $T_{\rm CMB} = 2.7255$ K and $N_{\rm eff} = 3.044$ (massless-neutrino approximation, recorded as a limitation) and $\Omega_{\rm DE} = 1 - \Omega_m - \Omega_k - \Omega_r$ by closure. Baseline parametrization CPL, $w(a) = w_0 + w_a(1-a)$; frozen comparison set for the parametrization axis: BA, $w(z) = w_0 + w_a\, z/(1+z^2)$ [36]; JBP, $w(z) = w_0 + w_a\, z/(1+z)^2$ [37]; BIN4, constant w in four redshift bins $z \in [0, 0.3)$, $[0.3, 0.7)$, $[0.7, 1.5)$, $[1.5, z_{\rm CMB}]$, with the future ($a > 1$) inheriting the first bin; and a Gaussian-process reconstruction, registered as a secondary cross-check (realization and verdict in Sec. VII D).

Priors. P1 (baseline, flat): $\Omega_m \in [0.01, 0.99]$; $H_0 \in [20, 100]$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; $\Omega_b h^2 \in [0.005, 0.1]$ when the CMB prior is used, else Gaussian 0.02218 ± 0.00055 (BBN, Ref. [24]); $\Omega_k = 0$ (baseline); $w_0 \in [-3, 1]$; $w_a \in [-3, 2]$ with $w_0 + w_a < 0$ (early matter domination); SN absolute magnitude $M_B \in [-20, -18]$, anchored by SH0ES when included, analytically marginalized otherwise. P2 (quintessence): P1 plus the pointwise condition $w(a) \geq -1$ for all $a \in [a_{\rm CMB}, a_{\rm max}]$. P3 (thawing): P2 plus $w_a \leq 0$ and $w_0 \geq -1$ (a phenomenological convention, so labeled).

Data combinations (frozen). D0 = DESI DR2 BAO + CMB distance prior + Pantheon+&SH0ES (baseline); D1 = D0 − SH0ES; D2 = D0 − CMB (BBN prior instead); D3 = D0 − BAO; D4 = D0 − SN; D5 (optional extension) = D0 with the SN compilation swapped.

Fate criteria. As given in Sec. IV, with numerical thresholds frozen in Appendix C, the priority order CRUNCH > RIP > DS > DECAY > OTHER, the $\times 2/\div 2$ boundary flag, the 1% OTHER audit budget whose breach triggers the amendment process, and the frozen thermodynamic mapping (DS, DECAY → heat-death-compatible; RIP, CRUNCH → non-heat-death).

Statistical rules. MCMC via `cobaya`; evidence via nested sampling; convergence $R - 1 < 0.01$ on the worst parameter; nested-model significance via Wilks $\Delta\chi^2$ with explicit p-values; model comparison via ΔAIC, ΔBIC, and $\ln B$; every fate probability reported with batch-means Monte Carlo error and boundary-sample fraction; any tail probability below 1% must be verified by nested sampling before entering the main text.

Fragility measures. $F_{\rm prior}$, $F_{\rm param}$, $F_{\rm data}$: the $\log_{10}$ span of $P(\textsc{rip})$ and $P(\text{heat-death})$ across {P1, P2, P3}, {CPL, BA, JBP, BIN4, GP}, and {D0–D4} respectively, each with all other choices fixed; a 100-mock null calibration at the ΛCDM best fit to D0; and the constraint horizon a_h, the scale factor at which the posterior-to-prior KL divergence of $w(a)$ first drops below 0.1 nat (BIN4).

Gates. Gate 1 (pipeline validation, blocking): reproduce the published DESI DR2 w_0w_aCDM result—best fit inside the published 68% contour and ΛCDM-exclusion significance within 0.3σ of the published $\approx 2.8\sigma$ (amended by A-001). Gate 2 (classifier calibration): $\geq 90\%$ of ΛCDM-truth mocks classified heat-death-compatible, and OTHER below 1% (amended by A-002). Gate 3: all chains entering the paper satisfy the convergence rule.

Freeze discipline. After the `git` tag, the plan text is immutable; every change is an append-only amendment entry recording what changed, why, and whether the trigger arose before or after real-data contact; exploratory (unplanned) analyses are permitted but must be labeled as such and kept separate from the preregistered core results.

Appendix B: Amendment log

The three amendments to the frozen plan are reproduced here in translation; the originals, with sign-off records, are in the repository. A-004–A-007 are listed after them as explicitly post-hoc records, not as additions to the preregistered core analysis.

A-001 (2026-07-03; during Gate 1, before any fate analysis). **Change:** the Gate-1 significance clause, "within 0.3σ of the literature value ($\approx 2.8\sigma$)," replaced by "consistent with published results using a *compressed* CMB likelihood of the same class" (Ref. [3] reports $N_\sigma \approx 1.1$–2.3 for DESI DR2 + compressed CMB + SN). The best-fit-point clause was unchanged. **Reason:** 2.8σ is a full-*Planck*(+lensing) number; a three-number distance prior discards lensing and spectral-shape information and cannot reach it by construction—an information deficit, not a pipeline error. Evidence: best fit $(w_0, w_a) = (-0.854, -0.520)$ against the official $(-0.838, -0.62)$, inside the 68% contour; CMB-prior $\chi^2 = 1.66$ at the *Planck* ΛCDM mean; our 2.25σ falls inside the published compressed-likelihood range. **Consequence:** every significance in this paper inherits the compressed-CMB condition (priced in Secs. III and VII).

A-002 (2026-07-06; predicted during the mock campaign from the probability-integral-transform argument, confirmed on completion). **Change:** the Gate-2 criterion "$\geq$ 90% of ΛCDM mocks judged heat-death-compatible" replaced by "the calibration deliverable is the null distribution itself; real-data fate statements must be reported in null-calibrated form." The OTHER and boundary budgets were unchanged (and were met). **Reason:** the criterion encoded a false factual presupposition. The ΛCDM truth sits on the zero-width RIP/DECAY boundary, so the heat-death-compatible posterior mass is distributed approximately as $U(0,1)$ under the null and *no* calibrated Bayesian procedure can exceed ∼50% direction accuracy (Sec. VI). Measured: 50.0% accuracy, KS $p = 0.25$ against uniformity, zero-residual mock at 52/48, all budgets met, six mid-campaign logged predictions correct—the failure mode matched the mathematical prediction, excluding a pipeline-defect explanation. **Consequence:** Gate 2's output became a core result of the paper rather than a pass/fail check. The amendment's "no calibrated procedure" wording is scoped in the revised analysis to the regular continuous CPL decision problem; it does not include priors with a discrete ΛCDM component or model averaging (Sec. VI).

A-003 (2026-07-06; triggered automatically by the frozen 1% OTHER *audit during the first parametrization-axis pass).* **Superseded by A-005; the formula below is preserved for provenance and is not used in the current results. Change:** an asymptotic branch added to the fate criteria: for parametrizations whose $w(a)$ has a finite limit w_∞ as $a \to \infty$, classify analytically—RIP iff $w_\infty < -1 - \epsilon$, DS iff $|w_\infty + 1| \leq \epsilon$ with persistent dark energy, DECAY iff $w_\infty > -1 + \epsilon$ ($\epsilon = 0.01$, matching the frozen DS threshold; boundary flagging unchanged). Divergent-w parametrizations (CPL, JBP) keep the original numerical criteria. **Reason:** measured OTHER fractions of 10.5% (BA) and 82.4% (BIN4) breached the budget: bounded or frozen futures fall into a gap between the numerical criteria (decay slower than the DECAY threshold, deviation larger than the DS threshold, power-law rips converging more slowly than the numerical RIP test at $a_{\max}$). The analytic criterion is exact in these cases (a constant $w < -1$ future implies a finite-time singularity). **Consequence for existing results: zero.** CPL and JBP had OTHER $= 0$; D0–D4, P2/P3, and the entire mock calibration are CPL-based and numerically unchanged. Reclassification affected BA and BIN4 only (Table IV).

A-004 (2026-07-21; post-hoc structural audit after all phase-3 results). The candidate common object was the seven-vector $w(a)$ at $a = (0.5, 0.67, 0.8, 1, 1.5, 2, 4)$. CPL, JBP, and BA have two-dimensional pushforward supports in that ambient space; BIN4 has a three-dimensional support because this grid maps to $(w_3, w_2, w_1, w_1, w_1, w_1, w_1)$ and does not sample w_4. Their common intersection is the one-dimensional set of constant histories, which has probability zero under each continuous native prior. Consequently, no nondegenerate common target is dominated by all four pushforwards and the required importance-density ratios do not exist. An earlier covariance-regularized Gaussian pilot was rejected because it manufactured full-dimensional densities and artificial overlap. The audit is a global No-Go: matched fate probabilities, ESS, and overlap are withheld. It changes no registered analysis or result.

A-005 (2026-07-21; corrective post-hoc amendment). Post-release review found that A-003 had used $\epsilon = 0.01$ both as a boundary tolerance and as the width of a de Sitter fate class. This is physically incorrect: every positive-dark-energy finite-limit future with $w_\infty < -1$ reaches a Big Rip in finite proper time, while exact de Sitter asymptotics occupy $w_\infty = -1$. The finite-limit branch is therefore corrected to RIP iff $w_\infty < -1$, DS iff $w_\infty = -1$, and DECAY iff $w_\infty > -1$; the registered ϵ is retained only as a boundary-adjacent flag. BA and BIN4 are reclassified in Table IV. CPL, JBP, D0–D4, P2/P3, null mocks, Wilks statistics, and Bayes evidence are unchanged. The original A-003 text remains above as provenance rather than being silently rewritten.

A-006 (2026-07-21; post-hoc robustness and scope amendment). The statement that $w_4 < 0$ keeps early dark energy subdominant was removed: it is not sufficient when w_4 lies just below zero. Future BIN4 fits impose $\rho_{\rm DE}/\rho_m(z = 1059) < 0.01$. A weighted audit of the archived posterior finds zero violating mass and a maximum sampled ratio 8.62×10^{-5}, so no refit or fate-probability change is required. A full-Planck run is a documented No-Go for the current background-only theory interface, which does not provide CMB spectra or perturbations. BIN4 results are accordingly restricted to the compressed-CMB-conditioned scope.

A-007 (2026-07-22; post-hoc implementation-provenance clarification). The frozen P1 table registered an early-matter-domination intent and wrote its CPL form, $w_0 + w_a < 0$, but did not spell out the translation for every grammar. The archived F-parametrization runs use $w_0 + w_a < 0$ for CPL, $w_0 < 0$ for JBP and BA, and $w_4 < 0$ for BIN4, corresponding respectively to each grammar's early-time equation of state. The translation note and the first F-parametrization outputs were first archived together in commit `777816f`; the current history therefore does not independently establish their within-commit order. This record documents the implementation without retroactively calling that detail explicitly preregistered. A-006 later strengthened the BIN4 condition to $\rho_{\rm DE}/\rho_m(z=1059)<0.01$. No chain, prior support, classifier, or reported probability changes under A-007.

Appendix C: Classifier numerics and known edges

The classifier integrates each posterior sample's background from $a=1$ to $a_{\max}=10^4$ and applies Sec. IV's criteria with the frozen thresholds: RIP tail-convergence tolerance 10^{-3} (relative change of $t(a)$ from $a_{\max}$ to $2a_{\max}$); DS tolerance $|w(a_{\max})+1|<0.01$; DECAY thresholds $\rho_{\rm DE}(a_{\max})/\rho_{\rm DE}(1)<0.01$ or $\rho_{\rm DE}/\rho_m(a_{\max})<1$; A-005 finite-limit boundary-adjacent tolerance $\epsilon=0.01$. Numerics: the age integral $\int da/(aH)$ is evaluated by adaptive quadrature on 40 logarithmically spaced panels; the CRUNCH scan uses a 400-point logarithmic grid in a; $\ln f_{\rm DE}$ is capped at 700 before exponentiation to avoid overflow (samples beyond the cap are already deep in the RIP regime). The boundary flag re-runs the classification with all thresholds doubled and halved and marks any sample whose label flips.

The implementation passes a ten-case unit suite: ΛCDM $\to$ DS (unflagged); phantom $w_a>0\to$ RIP; the DR2 best fit $\to$ DECAY; mild thawing $\to$ DECAY; a closed, decaying-DE case $\to$ CRUNCH; finite-limit cases just below, above, and exactly at $w_\infty=-1$ map to RIP, DECAY, and DS, respectively, with the boundary-adjacent flag raised; and two *documented limitation cases* of the frozen criteria, constant-w curves with $w=-0.98$ (quasi-Λ: neither DS-tight nor decayed at $a_{\max}$) and $w=-1.2$ (power-law phantom: a true finite-time singularity whose tail convergence at $a_{\max}=10^4$ is slower than the frozen tolerance), both of which are intentionally routed to OTHER, i.e. to the manual-audit channel. Constant-w curves are measure-zero in the CPL posterior; the posterior mass in the band $|w_a|<3\times10^{-4}$ is $\sim10^{-3}$, safely inside the 1% OTHER budget. These edges were documented before the real-data runs and did not require an amendment; the distinct gap later exposed by bounded-future grammars did (A-003, Appendix B).

For the GP construction the classifier receives $w_\infty=-1$ exactly: the conditional mean of a stationary kernel reverts to its mean function, so the asymptotic branch returns DS for every draw. The registered realization used six nodes at uniform $\ln a$ spacing over $z\in[0,3]$, a squared-exponential correlation, and mean $w = -1$. Its attempted hierarchical fit multiplied uniform node boxes by $\mathcal{N}(-1, \sigma_f^2 K_\ell)$, with $\sigma_f \in [0.01, 2]$ and $\ell \in [0.2, 3]$. That density was left unnormalized after box truncation, which can change both hyperparameter and node marginals, and the hierarchical chain stopped at $R - 1 = 0.232$. It therefore failed Gate 3 and no quantitative GP posterior is used. The sample-independent asymptotic construction remains valid because it follows from the assigned future mean, not from the nonconverged posterior. Separately, the cross-grammar machinery takes $r_{\rm drag}$ from a ΛCDM computation (drag-epoch physics treated as dark-energy independent); this is self-consistent a posteriori even for the binned grammar, whose free high-z bin sits at $w_4 = -1.71 \pm 0.62$. The A-006 weighted audit gives zero posterior mass above $\rho_{\rm DE}/\rho_m(z = 1059) = 0.01$ and a maximum sampled ratio 8.62×10^{-5}; this posterior check does not turn the compressed distance prior into a full-Planck likelihood.

Appendix D: Mock machinery and zero-residual validation

The null-calibration truth is the ΛCDM best fit to D0 ($\Omega_m = 0.2968$, $H_0 = 68.81$, $M_B = -19.397$), located by repeated bounded minimization. Each of the 100 mocks draws one noise realization per likelihood from the exact covariances used in inference—SN 1657×1657 (stat+sys, after the $z > 0.01$ mask), BAO 13×13, CMB prior 3×3—added to the truth's predicted data vectors, and is then processed by the full CPL+P1 pipeline (MCMC and classifier identical to the real-data runs). The campaign is a point null—mocks are generated at the single best-fit truth rather than marginalized over the ΛCDM posterior. For the direction layer this is sufficient: the uniformity prediction is truth-value-independent for any truth on the fate boundary (Appendix E), and every ΛCDM parameter point sits on it; the depth-layer floor inherits the usual parametric-bootstrap caveat of being conditioned on the fitted truth. Mocks are generated deterministically from a single seed, so the entire campaign regenerates bit-identically from the repository. A 101st, zero-residual realization (`mock000`) provides a closed-loop test: at the truth point all three likelihood χ^2 vanish identically, verifying the mock–likelihood round trip; its fate verdict (52/48) is the finite-covariance boundary diagnostic of Sec. VI.

Six falsifiable predictions were logged mid-campaign (after ∼17 of 100 mocks had been inspected), derived from the probability-integral-transform argument, before the campaign's completion and before the zero-residual realization ran: direction accuracy $50 \pm 10\%$; $P_{\rm heat}$ consistent with $U(0,1)$; 0–3 of 100 mocks below the real data's $P(\textsc{rip})$; OTHER and boundary budgets at the 10^{-3} level; `mock000` near 0.5; and the resulting A-002 trigger. All six were correct. Per-mock records (101 rows), summary statistics, and the batch logs are in the repository.

Appendix E: Boundary calibration statistics

This appendix collects, in an idealized regular-model form, the statistical facts behind the null calibration of Sec. VI: the exact uniformity result, its closed-form verification, the diagnostics that detect a *mis*calibrated pipeline, the finite-mock reporting rules, and the relation to the calibration literature.

One dimension. Let $\hat{x} \sim \mathcal{N}(\theta_0, \sigma^2)$ with posterior $\theta \,|\, \hat{x} \sim \mathcal{N}(\hat{x}, \sigma^2)$ (the calibrated case), and let class A be $\theta > 0$ with the truth on the boundary, $\theta_0 = 0$. Then $P(A \,|\, \hat{x}) = \Phi(\hat{x}/\sigma) = \Phi(Z)$ with $Z \sim \mathcal{N}(0,1)$: by the probability integral transform [38], the class probability is *exactly* $U(0,1)$ across realizations. Numerically (2×10^5 replications): mean 0.5010, variance 0.08356 against the uniform value $1/12 \approx 0.08333$, direction rate 0.502, KS $p = 0.16$. Because the class probability is a continuous functional of the data, its null distribution has no atom at 1/2: individual realizations can look arbitrarily confident while the ensemble verdict remains a fair coin.

Two dimensions, correlated errors. In a (u, v) analogue of $(w_0{+}1, w_a)$ with a D0-like degenerate covariance and a linear fate score $s = \alpha u + v$ ($\alpha = 1.2$, class $s > 0$), the same argument applies to any linear boundary through the truth: 5×10^4 replications give mean 0.5013, variance 0.08326, direction rate 0.505, KS $p = 0.058$ (at 5×10^4 replications the KS test resolves sub-percent distortions; this is comfortable agreement). The CPL fate boundary $w_a = 0$ is exactly linear, so the result covers the geometry of Sec. VI up to the non-Gaussianity of the real posterior—which the full-pipeline campaign tests directly. The derivation assumes a continuous posterior with no atom on the boundary. It does not apply unchanged to spike-and-slab priors or Bayesian model averaging that assigns positive probability to exact ΛCDM.

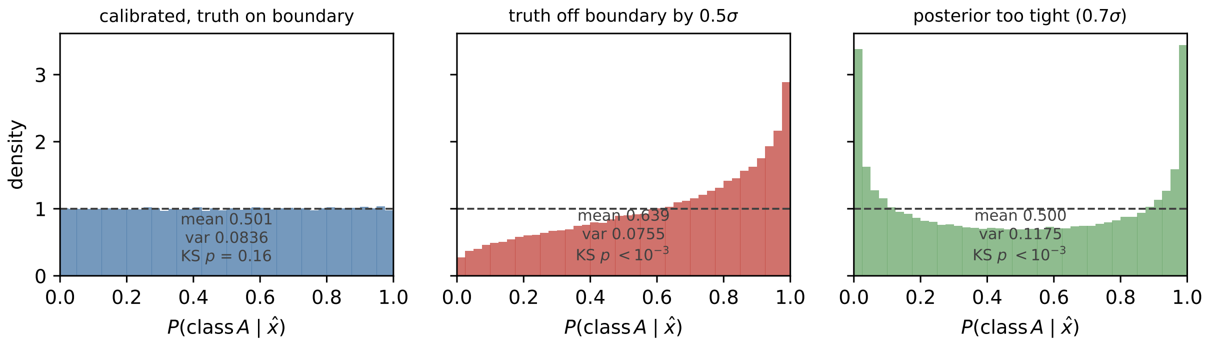

FIG. 7. The boundary-calibration toy (Appendix E). Left: calibrated posterior with the truth on the boundary—class probabilities exactly uniform. Center: truth off the boundary by 0.5σ—direction information appears, as it should. Right: over-confident posterior—mean preserved but variance above $1/12$; the uniformity diagnostics fail precisely when the pipeline is defective.

Diagnostics: when uniformity must fail. The value of the null test lies in what breaks it (Fig. 7). Truth off the boundary by 0.5σ: mean 0.639, direction rate 0.692, KS $p \approx 0$—direction information appears exactly when it exists. Decision threshold shifted while the truth sits on it: still uniform (KS $p = 0.25$)—only the truth–boundary distance matters. Posterior too wide (under-confident, 1.5σ): mean stays at 0.500 but the variance collapses to $0.0496 < 1/12$. Posterior too tight (over-confident, 0.7σ): variance inflates to $0.1175 > 1/12$. Strongly curved boundary: KS $p = 2.5 \times 10^{-6}$—uniformity is a locally-linear statement (the CPL boundary is exactly linear). The triple observed in Sec. VI—accuracy $\approx 50\%$, uniform shape, variance consistent with $1/12$—shows no gross defect of the tested kinds. The diagnostics respond in distinct directions in the simulations, but a 100-realization campaign retains limited power against mild misspecification.

Finite-mock reporting rules. With k of n fitted-truth mocks at or below the real-data statistic, we report the conditional plus-one estimate $\hat{p} = (k+1)/(n+1)$ [39] and, for $k = 0$, the 95% upper bound $1 - 0.05^{1/n}$; a Monte Carlo p-value is never reported as zero. These standard rules were adopted during the campaign and applied uniformly; they weaken, not strengthen, the headline bound. For $k = 0$, $n = 100$: $\hat{p} = 0.0099$, upper bound 0.0295—quoted throughout as $p < 0.030$.

Relation to the calibration literature. The lineage runs from the probability integral transform [38] through prequential calibration [40] and posterior predictive checking [41, 42].

The closest modern relative is simulation-based calibration [43], which validates posterior computation via rank uniformity under prior-drawn truths. The present test differs in both conditioning and target: the truth is *fixed at the boundary point*, and the calibrated object is a *decision functional discontinuous there*—the campaign is a null test of the verdict, not of the posterior machinery (which Gate 1 validates separately). The practical residue is a reporting discipline matching the three layers of Sec. VIII: a fate-style claim should state (i) its direction-layer verdict together with the null distribution of that verdict, (ii) its depth-layer tail probability under the finite-simulation rules above, and (iii) the translation-layer conditions—parametrization, prior, statistic—under which the verdict holds; Table V is the translation layer's error budget.

Appendix F: Reproducibility

The repository, `https://github.com/shlinawaf717-collab/cosmo_fate` (archived at Zenodo, `https://doi.org/10.5281/zenodo.21332433`), contains the frozen plan and amendment log, the full pipeline (likelihood configurations, fate classifier and its unit suite, mock generator, batch drivers, nested verification), archived thin chains and per-mock records sufficient for the reported summaries, the figure scripts, and deterministic audits for induced prior fate mass, in-window chain minima, model averaging, the registered constraint-horizon metric, the A-004 matched-prior structural No-Go, and the A-007 grammar-prior implementation trace. All input data are public: DESI DR2 BAO compressed measurements and covariance; Pantheon+ (with SH0ES), Union3, and DES-Dovekie supernova files obtained via the `cobaya` installer; the CMB distance prior of Ref. [19] (Table I, ΛCDM row, *Planck* 2018 TT,TE,EE+lowE), entered by value; and the official DESI DR2 w_0w_aCDM chains used in Gate 1, downloaded from the public DESI data portal. Every input file is registered in the repository manifest with source, version, and SHA-256 checksum.

Software: Python 3.13, `cobaya` 3.6.2, `camb` 1.6.6, `dynesty` 3.0.0, `getdist` 1.7.7, `numpy` 2.5.0, `scipy` pinned to 1.16.2 (a scipy 1.18 regression in spline evaluation breaks `camb` 1.6.6's BBN interface; the pin and the upstream fix are documented in the manifest). The classifier unit suite, the zero-residual closed-loop test, deterministic mock regeneration, and CI audit invariants together allow every number in this paper to be reproduced from the repository; the environment setup, data installation, and per-result commands are documented in the repository README.

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